

Qn	Suggested Mark Scheme	Mark																														
1(a)(i)	Tabulates <u>mass of weighing bottle</u> and <u>mass of weighing bottle and residual FA 3</u> for table and <u>mass of FA 3 used</u>. Table has <u>appropriate headers</u> and <u>units</u>. Tabulation may be <u>vertical or horizontal</u>; lines are not essential but there should be no absences of headers. Where units have not been included in the header, there should be the <u>appropriate unit</u> for each entry in the table. <u>Weighing</u> <table><tr><td>Mass of (empty) weighing bottle</td><td>/ g</td><td></td></tr><tr><td>Mass of weighing bottle and FA 3</td><td>/ g</td><td></td></tr><tr><td>Mass of weighing bottle and residual FA 3</td><td>/ g</td><td></td></tr><tr><td>Mass of FA 3 used</td><td>/ g</td><td></td></tr></table>	Mass of (empty) weighing bottle	/ g		Mass of weighing bottle and FA 3	/ g		Mass of weighing bottle and residual FA 3	/ g		Mass of FA 3 used	/ g		1																		
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Mass of FA 3 used	/ g																															
	Correctly calculates the mass of FA 3 used is <u>within 9.8 to 10.2 g</u> , both inclusive.	1																														
(ii)	All the final/initial <u>burette readings</u> for <u>all accurate titres</u> are recorded to the <u>nearest 0.05 cm³</u>. <i>Treat all titres as “accurate” unless labelled rough or 1st titre is of lower precision than subsequent titres</i> Table has <u>appropriate headers</u> and <u>units</u>. Tabulation may be <u>vertical or horizontal</u>; lines are not essential but there should be no absences of headers. Where units have not been included in the header, there should be the <u>appropriate unit</u> for each entry in the table. <u>Titration</u> <table><tr><td>titration number</td><td></td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>volume of FA 4 added</td><td>/ cm³</td><td>6.00</td><td>18.00</td><td>10.00</td><td>14.00</td></tr><tr><td>initial burette reading</td><td>/ cm³</td><td></td><td></td><td></td><td></td></tr><tr><td>final burette reading</td><td>/ cm³</td><td></td><td></td><td></td><td></td></tr><tr><td>volume of FA 2 used</td><td>/ cm³</td><td></td><td></td><td></td><td></td></tr></table>	titration number		1	2	3	4	volume of FA 4 added	/ cm ³	6.00	18.00	10.00	14.00	initial burette reading	/ cm ³					final burette reading	/ cm ³					volume of FA 2 used	/ cm ³					1
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	Proposes two extra volumes* of FA 4 which are within the range 6 – 18 cm³. * gaps between adjacent FA 4 volumes should be $\geq 3.5 \text{ cm}^3$ and $\leq 4.5 \text{ cm}^3$	1																														

(b)	Plotting – all points within $\pm\frac{1}{2}$ small square. Check all points and put tick at each point if correct.	1
	The graph line is the <u>straight</u> best-fit line and line is correctly extrapolated to intersect both axes. i.e. one continuous straight line and no point is further than 1 cm ³ away from the line in <u>either direction</u> .	1
	<u>Accuracy mark</u> scaled $V_{\max}(\text{FA 4}) = \text{graphical value of } V_{\max}(\text{FA 4}) \times \frac{\text{mass of FA 3 used}}{10.00}$ Award this mark if the difference between the student's <u>scaled</u> $V_{\max}(\text{FA 4})$, and the teacher's <u>scaled</u> $V_{\max}(\text{FA 4})$ values is no greater than $\pm 1.5 \text{ cm}^3$.	1
	<u>Accuracy mark</u> The value of $V_{\max}(\text{FA 2})$ is independent of the concentration of FA 4, so no scaling is needed for this mark. Award this mark if the difference between the student's, and the teacher's, graphical $V_{\max}(\text{FA 2})$ values is no greater than $\pm 1.0 \text{ cm}^3$.	1
	Student recognises that $V_{\max}(\text{FA 2})$ and $V_{\max}(\text{FA 4})$ are the intercept values on the y-axis and the x-axis respectively and these values are correctly read from the graph to $\pm\frac{1}{2}$ small square (ignore sig figs) If the student has not shown the extrapolation on the graph, the student may still earn this mark if the values quoted correspond to correct extrapolation of the student's graph line.	1
(c)	Use clearly stated co-ordinates <u>from graph</u> . <i>Coordinates could be from $V_{\max}$ values.</i> or Correct values of Δtitre and of $\Delta(\text{volume of FA 4 used})$ obtained from a <u>clearly drawn triangle</u> which covers at least <u>3 large squares</u> in each direction (<i>allow measurements to $\pm \frac{1}{2}$ small square</i>). and - Correctly calculate gradient <i>with negative sign</i>. (ignore sig figs for this mark) Do not award this mark if <i>the graph line is not straight.</i> <i>For this mark, ignore missing working and incorrect/missing units.</i>	1

	Minimal: Student used $V_{\max}$ (FA 4) and $V_{\max}$ (FA 2) to determine gradient without working or labeling of coordinates.	
(d)	Negative gradient because the more Na_2CO_3 / FA 4 added, the less NaOH / FA 2 is needed for neutralisation (or words to that effect) (owtte) - slope of graph has a <u>negative gradient</u> - as <u>more Na_2CO_3</u> are added, amount of H_2SO_4 partially neutralised by it will increase and - hence <u>less NaOH</u> is required to neutralise the remaining H_2SO_4	1
(e)(i)	Correctly calculates $[\text{Na}_2\text{CO}_3]$ in FA 4. $[\text{Na}_2\text{CO}_3] = \frac{\text{mass of FA 3 used}}{106.0} \times \frac{1000}{250} = A \text{ (mol dm}^{-3}\text{)}$ <i>Do NOT award this mark if the mass of FA 3 was incorrectly calculated in the table in part (a).</i>	1
(e)(ii)	Student chooses to use the $V_{\max}$(FA 4) value and for using it effectively and correctly calculates the $n\text{Na}_2\text{CO}_3$ it contains $n\text{Na}_2\text{CO}_3 = \frac{V_{\max}(\text{FA 4})}{1000} \times A = B \text{ (mol)}$ $[\text{H}_2\text{SO}_4] = B \times \frac{1000}{25.0} = C \text{ (mol dm}^{-3}\text{)}$ <i>Allow both marks for a <u>correct answer</u>, even if the working is missing. The 1st mark is for choosing to use the student's recorded value for $V_{\max}$(FA 4) and for using it effectively.</i> <i>Note: Students who do not calculate $n\text{Na}_2\text{CO}_3$, but who incorporate all the appropriate figures in an expression which yields a correct value for C, should not be penalised under the 'shows working' mark because of this.</i>	1 1
(e)(iii)	Using $V_{\max}$(FA 2) to calculate $[\text{NaOH}]$ Using $[\text{H}_2\text{SO}_4] = C$ $[\text{NaOH}] = 2C \times \frac{25.0}{V_{\max}(\text{FA 2})} = D \text{ (mol dm}^{-3}\text{)}$ Or Using $n\text{Na}_2\text{CO}_3$, B $[\text{NaOH}] = 2B \times \frac{1000}{V_{\max}(\text{FA 2})} = D \text{ (mol dm}^{-3}\text{)}$	1
(f)	Correctly calculates $[\text{NaOH}]$ using the given expression.	1

(g)	effect: value of $V_{\max}(\text{FA 4})$ will be higher <i>than expected</i> / greater / increase / too high explanation: presence of water means <u>mass of Na_2CO_3 weighed out is lower than expected</u>, so <u>$[\text{Na}_2\text{CO}_3]$ in FA 4 is lower than expected</u>, so more FA 4 is needed to neutralise the sulfuric acid. Cannot accept: mass of FA 3 weighed out is lower than expected (because mass of FA 3 includes the water)	1
	effect: value of $V_{\max}(\text{FA 2})$ is <i>same as expected</i> / unchanged / no effect / remains the same explanation: the <u>volume of FA 4 added at $V_{\max}(\text{FA 2})$ is zero</u>, so the incorrect concentration of FA 4 has no effect.	1

Teacher's values

Date	Shift	Teacher's value			
		$V_{\max}(\text{FA4})$	mass	scaled	$V_{\max}(\text{FA2})$
28/8/2018	1 (0800 – 1030)	25.50	9.958	25.39	35.50 (CL1-4) 35.00 (CL5, PL1 – 3)
	2 (1110 – 1340)	25.50 (all labs) / 26.25* (CL5, PL1–3)	10.045	25.61 (all labs) / 26.37* (CL5, PL1 – 3)	35.50 (CL1-4) 35.00 (CL5, PL1–3)
	3 (1420 – 1650)	25.50 (all labs) / 26.25* (CL5, PL1–3)	10.045	25.61 (all labs) / 26.37* (CL5, PL1–3)	35.50 (CL1-4) 35.00 (CL5, PL1–3)

Note: $V_{\max}(\text{FA2})$ differ for level 2 and level 3 labs due to different stock solutions used.

Student's/Teacher's scaled $V_{\max}(\text{FA 4}) = \text{graphical value of } V_{\max}(\text{FA 4}) \times \frac{\text{mass of FA 3 used}}{10.00}$

Qn	Suggested Mark Scheme	Mark												
2(a), (b)	All <u>measured temperatures</u> (initial temperature of FA 5, FA 6 and $T_{\max}$ after reaction) are recorded to <u>0.1 °C</u> and Correctly calculate T_{av} and $\Delta T_{\max}$ (to 0.1 °C or up to 2 d.p.) using the two expressions below. $T_{\text{av}} = \frac{(\text{volume FA 5} \times \text{initial temp FA 5}) + (\text{volume FA 6} \times \text{initial temp FA 6})}{\text{total volume of reaction mixture}}$ $\Delta T_{\max} = T_{\max} - T_{\text{av}}$	1												
	All <u>measured temperatures</u> (initial temperature of FA 5, FA 7 and $T_{\max}$ after reaction) are recorded to <u>0.1 °C</u> and Correctly calculate T_{av} and $\Delta T_{\max}$ (up to 2 d.p.) using the two expressions below. $T_{\text{av}} = \frac{(\text{volume FA 5} \times \text{initial temp FA 5}) + (\text{volume FA 7} \times \text{initial temp FA 7})}{\text{total volume of reaction mixture}}$ $\Delta T_{\max} = T_{\max} - T_{\text{av}}$	1												
	<u>Accuracy mark</u> If ΔT for both experiments ± 0.2 from Teacher's result If ΔT for both experiments ± 0.6 from Teacher's result <i>Teacher's results</i> <table border="1"> <thead> <tr> <th>Shift</th><th>ΔT for FA 6</th><th>ΔT for FA 7</th></tr> </thead> <tbody> <tr> <td>1</td><td>7.3</td><td>1.3</td></tr> <tr> <td>2</td><td>7.2</td><td>1.3</td></tr> <tr> <td>3</td><td>6.8</td><td>1.1</td></tr> </tbody> </table>	Shift	ΔT for FA 6	ΔT for FA 7	1	7.3	1.3	2	7.2	1.3	3	6.8	1.1	2 (max) 1
Shift	ΔT for FA 6	ΔT for FA 7												
1	7.3	1.3												
2	7.2	1.3												
3	6.8	1.1												
(c)(i)	Correctly calculates $n\text{HCl} = \frac{40.0}{1000} \times 2.00 = 0.0800 \text{ mol}$ and $n\text{NaOH} = \frac{50.0}{1000} \times 1.00 = 0.0500 \text{ mol}$ 0.0500 mol NaOH reacts with 0.0500 mol HCl. Thus, FA 5 (or HCl) is in excess.	1												

(c)(ii)	Correctly calculates $q = (V_{FA\ 5} + V_{FA\ 6})(1.00) c\Delta T$ $= 90 \times 1.00 \times 4.18 \times \Delta T_{\max}$ J (allow kJ) Ignore if q has sign or has a wrong unit or if $\Delta T_{\max}$ was calculated wrongly.	1
	Recognises that the limiting reagent is HCl and uses nHCl of to determine ΔH_1 . Correctly calculates $\Delta H_1 = -\frac{q}{n_{HCl}} = \underline{-a}$ kJ mol ⁻¹ Correct sign for $\Delta H_1 = -ve$	1
(d)(i)	Correctly calculates $n_{HCl} = \frac{40.0}{1000} \times 2.00 = 0.0800$ mol and $n_{Na_2CO_3} = \frac{50.0}{1000} \times 1.00 = 0.0500$ mol 0.0500 mol Na ₂ CO ₃ requires 0.100 mol HCl OR 0.0400 mol Na ₂ CO ₃ reacts with 0.0800 mol HCl Thus, FA 7 (or Na ₂ CO ₃) is in excess.	1
(d)(ii)	Correctly calculates $q = (V_{FA\ 5} + V_{FA\ 7})(1.00) c\Delta T$ $= 90 \times 1.00 \times 4.18 \times \Delta T_{\max}$ J (allow kJ) Ignore if q has sign or has a wrong unit or if ΔT was calculated wrongly.	1
	Recognises that the limiting reagent is HCl and uses nHCl of to determine ΔH_2 . Correctly calculates $\Delta H_2 = -\frac{q}{n_{HCl}} \times 2 = \underline{-b}$ kJ mol ⁻¹ Correct sign for $\Delta H_2 = -ve$	1
(e)	Correctly calculates ΔH_3 $ \begin{array}{ccc} 2NaOH(aq) + CO_2(g) & \xrightarrow{\Delta H_3} & Na_2CO_3(aq) + H_2O(l) \\ \downarrow 2 \times \Delta H_1 & & \downarrow \Delta H_2 \\ & & 2NaCl(aq) + CO_2(g) + 2H_2O(g) \end{array} $ By Hess' Law, $\Delta H_3 = 2\Delta H_1 - \Delta H_2$ $= 2(-a) - (-b)$ $= \underline{b - 2a}$ kJ mol ⁻¹	2
	Shows working in all calculations in 1(c), (e) – (f) and 2(c) – (d) . All calculations must be relevant although they may not be complete or correct. Any calculations not attempted loses this mark.	1
	Shows appropriate significant figures (3 s.f.) in all final answers in 1(c), (e) – (f) and 2(c)(ii), (d)(ii) . Shows appropriate units in 1(e) – (f) and 2(c) – (d) . Any calculations not attempted loses this mark.	1

(f)	Recognise that enthalpy change of solution of solid Na_2CO_3 , $\Delta H_{\text{solution}}(\text{Na}_2\text{CO}_3)$, is required.	1									
	Suitable energy cycle or working to show how the 3 enthalpy change terms can be used to determine ΔH_4 $ \begin{array}{ccc} 2\text{NaOH}(\text{aq}) + \text{CO}_2(\text{g}) & \xrightarrow{\Delta H_4} & \text{Na}_2\text{CO}_3(\text{s}) + \text{H}_2\text{O}(\text{l}) \\ \downarrow 2 \times \Delta H_1 & & \downarrow \Delta H_{\text{solution}}(\text{Na}_2\text{CO}_3) \\ 2\text{NaCl}(\text{aq}) + \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g}) & \xleftarrow{\Delta H_2} & \text{Na}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \end{array} $ By Hess' Law, $\Delta H_3 = 2\Delta H_1 - \Delta H_2 - \Delta H_{\text{solution}}(\text{Na}_2\text{CO}_3)$ $= \mathbf{b - 2a - \Delta H_{\text{solution}}(\text{Na}_2\text{CO}_3)}$ kJ mol^{-1}										
	<u>Suggested plan</u> To determine $\Delta H_{\text{solution}}(\text{Na}_2\text{CO}_3)$ 1. Weigh a <u>fixed/known mass</u> (accurately about 5.0 g) of solid Na_2CO_3 in a weighing bottle 2. Using a (50.0 cm^3) measuring cylinder, transfer <u>50.0 cm^3 of deionised water</u> into <u>Styrofoam cup</u>. (Place this cup into a second Styrofoam cup which is placed in a 250 cm^3 glass beaker). 3. Record the <u>initial temperature of the water</u>. 4. Add (rapidly) solid Na_2CO_3 to the water and replace the lid, <i>keeping the bulb of the thermometer in the solution throughout</i>. Stir the mixture. 5. Continue to stir the mixture and record the <u>maximum change from the initial temperature</u>. 6. <i>Reweigh</i> the weighing bottle. <table border="1" data-bbox="284 1350 1353 1912"> <thead> <tr> <th>Marks</th><th>Requirements</th><th>Key Marking Points</th></tr> </thead> <tbody> <tr> <td>[1]</td><td>Appropriate <u>quantities</u> of substances used and <u>appropriate apparatus</u> and their <u>capacities</u></td><td> • vol of water = <u>50 cm^3</u> • water : 50 cm^3 measuring cylinder or 50 cm^3 burette • mass of Na_2CO_3 = 5 g or any known / fixed value • Styrofoam cup with lid </td></tr> <tr> <td>[1]</td><td>Essential <u>experimental details</u> to ensure <u>reliable results</u></td><td> 1. Take <u>initial</u> and <u>maximum rise / drop in temperature</u>. 2. Start reaction by <u>adding Na_2CO_3</u> and rapidly and replace lid. 3. <u>Stir</u> the reaction mixture. 4. <u>Reweigh</u> the weighing bottle. (any 3 points) </td></tr> </tbody> </table>	Marks	Requirements	Key Marking Points	[1]	Appropriate <u>quantities</u> of substances used and <u>appropriate apparatus</u> and their <u>capacities</u>	• vol of water = <u>50 cm^3</u> • water : 50 cm^3 measuring cylinder or 50 cm^3 burette • mass of Na_2CO_3 = 5 g or any known / fixed value • Styrofoam cup with lid	[1]	Essential <u>experimental details</u> to ensure <u>reliable results</u>	1. Take <u>initial</u> and <u>maximum rise / drop in temperature</u>. 2. Start reaction by <u>adding Na_2CO_3</u> and rapidly and replace lid. 3. <u>Stir</u> the reaction mixture. 4. <u>Reweigh</u> the weighing bottle. (any 3 points)	2
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Qn	Suggested Mark Scheme		Mark	
3(a)	See answer grid <u>12 possible observations</u>		3	
	Marks allocated as follows:			
	Number of correct observations	Award (max 3)		
	10 – 12	3		
	7 – 9	2		
	4 – 6	1		
(b)(i)	FA 8	alcohol	gives negative <u>2,4–DNPH test</u> in (a)(i) but positive <u>iodoform test</u> in (a)(iv) shows that it contains $\text{CH}_3\text{CH}(\text{OH})\text{--}$ group and is an alcohol.	4
	FA 9	ketone	gives a positive <u>2,4–DNPH test</u> in (a)(i) but negative <u>Fehling’s test</u> in (a)(iii) shows that it is a ketone.	
	FA 10	carboxylic acid	gives H_2 on <u>reaction with Mg</u> in (a)(ii) shows that it contains the $\text{--CO}_2\text{H}$ group / is a carboxylic acid.	
	FA 11	aldehyde	positive <u>Fehling’s test</u> in (a)(iii) shows that it contains the --CHO group / is an aldehyde.	
	[1] each for quoting of evidence, can give either the name of the reagent or the qn for alcohol, stating the presence of $\text{CH}_3\text{CH}(\text{OH})\text{--}$ group is essential additional evidence (not required for identity to be deduced) is penalised			
(ii)	sample: FA 11 identity: <u>Cu_2O</u> type of reaction: <u>redox</u>		1	
(c)(i)	$\text{C}_6\text{H}_6 + \text{Br}_2 \longrightarrow \text{C}_6\text{H}_5\text{Br} + \text{HBr}$		1	
(ii)	amount of bromine = $\frac{6.0 \times 3.12}{159.8}$ = 0.1171 mol		1	
	amount of benzene = $\frac{0.1171}{2}$ = <u>0.005857 mol</u>			
	mass of benzene = 0.005857×78.0 = 4.568 g		1	
	volume of benzene = $\frac{4.568}{0.88}$ = <u>5.19 cm³</u>			

(iii)	Volume of benzene used is within the range of <u>5.19 – 20.00 cm³</u>. Volume of bromine is <u>kept unchanged</u>. To keep mole ratio of benzene : bromine within 2:1 and 1:2. Suggested answer: <table><tr><th>n(benzene):n(Br₂)</th><th>n(benzene) / mol</th><th>n(Br₂) / mol</th><th>V(benzene) / cm³</th><th>V(Br₂) / cm³</th></tr><tr><td>1:1</td><td>0.1171</td><td>0.1171</td><td>10.0</td><td>6.0</td></tr><tr><td>1:1.5</td><td>0.007806</td><td>0.1171</td><td>7.0</td><td>6.0</td></tr></table>	n(benzene):n(Br ₂)	n(benzene) / mol	n(Br ₂) / mol	V(benzene) / cm ³	V(Br ₂) / cm ³	1:1	0.1171	0.1171	10.0	6.0	1:1.5	0.007806	0.1171	7.0	6.0	1
n(benzene):n(Br ₂)	n(benzene) / mol	n(Br ₂) / mol	V(benzene) / cm ³	V(Br ₂) / cm ³													
1:1	0.1171	0.1171	10.0	6.0													
1:1.5	0.007806	0.1171	7.0	6.0													
(iv)	<u>Suggested Plan</u> <u>Preparation of crude bromobenzene</u> - Using a 20 cm³ measuring cylinder, measure 20.0 cm³ (0.226 mol) of benzene, and a 10 cm³ measuring cylinder to measure 6.0 cm³ (0.117 mol) of bromine. Add both into a 50 cm³ round-bottomed flask. - Add an iron tack to the reaction mixture. - Using a water bath, gently heat the flask fitted with a reflux condenser at 50 – 55 °C for about 15 minutes. Reaction should be done in a fume cupboard to remove the acidic fumes of hydrogen bromide evolved. - After about 15 min when the reaction subsides, remove the reaction mixture from the water bath. - The mixture is then transferred into a 100 cm³ conical flask heated over a hot plate to drive off unreacted bromine until the bromine vapour above the liquid disappears (or solution turns from brown/orange to colourless). This should be done in the fume cupboard to collect the bromine gas evolved. - Allow the mixture to stand and cool before doing solvent extraction. - Using a 25 cm³ measuring cylinder, add 25.0 cm³ of ether to dissolve the cooled reaction mixture. Open the cover to release the pressure build-up. - Using a 10 cm³ measuring cylinder, add 5 cm³ of sodium hydroxide to the ether solution before shaking the mixture in a 100 cm³ separatory funnel. Open the tap to release the aqueous layer. - Repeat step 8. - Using a 10 cm³ measuring cylinder, add 10 cm³ portion of water to the ether solution before shaking the mixture in the separatory funnel. Open the tap to release the aqueous layer. (Alternatively, use conical flask to shake the mixture in steps 8–10 and decant the top layer or use a dropper/ pipette to withdraw bottom aq layer) - Add a spatula full of magnesium sulfate to the ether layer, swirl and stand for a few minutes to dry the ether layer before it is filtered away. - Perform fractional distillation and collect the fraction between 140–160 °C for bromobenzene. - Weigh the bromobenzene collected. - Calculate the percentage yield of bromobenzene based on benzene. This is the yield. - Repeat the experiment with the sets of reagents in (iii) and determine the yield for these reaction mixtures.	6															

Marks	Requirements	Key Marking Points
[2]	Logical flow of plan	• mixing reagents based on given quantities, warm for 15 min at specified temperature • (heat to drive off bromine) • follow steps in solvent extraction • dry the crude sample using MgSO₄ and filter • fractional distillation • weigh the product obtained/measure volume collected • <u>Repeat</u> the experiment <u>two more times</u> using the volumes suggested in (iii). (1 mark for any 4 points, 2 marks for all points)
[1]	<u>Appropriate apparatus</u> and their <u>capacities</u>	• 50 cm³ round bottomed flask • 10 or 25 cm³ measuring cylinders • 100 cm³ separatory funnel / separating funnel
[2]	Essential <u>experimental details</u> to ensure <u>reliable results</u>	• Carrying out reaction <u>under reflux</u> to prevent loss of volatile reagents. • Ensure <u>unreacted Br₂ is driven off</u> by observing the <u>clearing of Br₂ vapour</u> above the reaction mixture. • <u>Collecting</u> the bromobenzene fraction between <u>140–160 °C</u> (or T between 100 and 200 °C to avoid collecting 1,4–dibromobenzene). (1 mark for any point, max. 2 marks)
[1]	Safety precautions	Carrying out the reaction in fumehood to remove the acidic HBr fume and irritant Br₂. Release the pressure from the separatory funnel occasionally. (1 mark for any point)
[1]	Calculations	Theoretical amount of bromobenzene based on benzene = n(benzene) $\% \text{ yield} = \frac{\text{actual amount of bromobenzene}}{\text{theoretical amount of bromobenzene}} \times 100\%$ % yield can also be in terms of mass or volume of bromobenzene. Should also show an outline of how the amount of bromobenzene are obtained from its mass or volume and/or vice versa. Note: The yield in this question is more of to determine the % conversion of benzene to bromobenzene in the given reaction mixtures and not comparing the % yield of the different reaction mixtures. The fixed reference of benzene given in the question is necessary due to changing limiting reagents in the varying proportions of benzene and bromine.
Maximum 6 marks awarded for the 7 marking points.		

(v)	Step 2: <u>electrophilic substitution</u>	1
	Step 3: <u>nucleophilic substitution</u>	1

Mark Scheme Grid for parts 3(a)(i) – 3(a)(iv)

(a)	test	observations with			
		FA 8	FA 9	FA 10	FA 11
(i)	Place about 1 cm depth of the unknown in a test-tube. To the test-tube, add about 1 cm depth of 2,4-dinitrophenylhydrazine reagent. Shake and warm the test-tube in a water-bath.	no observable change	orange / yellow ppt observed (1)	no observable change (1)	orange / yellow ppt observed (1)
(ii)	Place about 1 cm depth of the unknown in a test-tube. To the test-tube, add a small amount of magnesium powder.	no observable change (1)	no observable change	effervescence observed and gas gives a 'pop' sound with lighted splinter (1)	no observable change (1)
(iii)	Place about 1 cm depth of the unknown in a test-tube. To the test-tube, add 1 cm depth of Fehling's solution. Shake and warm the test-tube in a water-bath.	no observable change (1)	no observable change (1)	no observable change	brick-red ppt observed (1)
(iv)	Place about 1 cm depth of the unknown in a test-tube. To this test-tube, add about 3 cm depth of iodine solution. Then add sodium hydroxide dropwise until the iodine colour has almost discharged. Warm the test-tube in a water-bath if necessary.	yellow crystals / solid observed (1)	yellow crystals / solid observed (1)	no observable change (1)	no observable change